

Exercise Sheet Week 5

Key:

- Questions marked with a 🍷 are for discussion in your tutorials. Please complete these questions before your tutorial.
- Questions marked with a 🎵 have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🍷 **Exercise 1.** (i) Write down the vector equation of a plane.

(ii) Find an equation of a plane containing the points $A = (1, 1, 2)$, $B = (3, 4, 1)$, and $C = (-5, 1, -1)$.

(iii) What is the shortest distance from the origin to this plane?

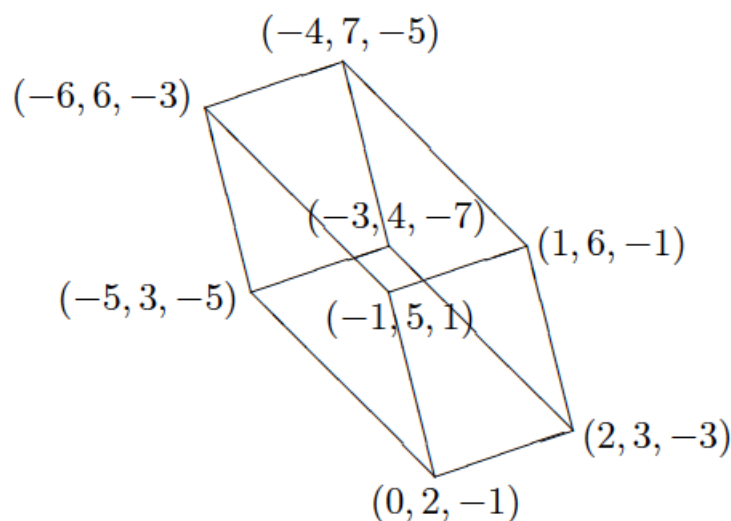
🍷 **Exercise 2.** Obtain a unit vector parallel to the y - z plane and perpendicular to $\vec{v} = \hat{i} + 4\hat{j} - 3\hat{k}$.

🎵 **Exercise 3.** Let $a = (1, 2)$, $b = (-1, 3)$ and $c = (0, 7)$ be three points. Let L_1 be a line through a and b , L_2 a line through c parallel to L_1 , and L_3 a line through c perpendicular to L_1 . Find

(i) the point of intersection of L_1 and L_3 .

(ii) the perpendicular distance between c and L_1 .

🎵 **Exercise 4.** Consider the parallelepiped whose corners are the points $(0, 2, -1)$, $(2, 3, -3)$, $(-1, 5, 1)$, $(-5, 3, -5)$, $(-6, 6, -3)$, $(-3, 4, -7)$, $(-4, 7, -5)$, $(1, 6, -1)$ as shown in the diagram below. What is the volume of this solid?



🎵 **Exercise 5.** Find the point at which the line with parametric equations $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intersects the plane $4x + 5y - 2z = 18$.